KG COLLEGE OF ARTS AND SCIENCE
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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

EMPLOYEE ATTENDANCE MONITORING AND LEAVE MANAGEMENT SYSTEM

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DECLARATION

I hereby declare that the project entitled "**Employee Attendance Monitoring and Leave Management System"** submitted in partial fulfillment of the requirements for the course **Java programming** is a **record of my original work** carried out under the guidance of **Mugilan R** Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

RAKSHAN D

CERTIFICATE

This is to certify that the project entitled **"Employee Attendance Monitoring and Leave Management System"** is a **bona fide work** carried out by **Rakshan D ,Information Technology**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java programming**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date:

Submitted for the Viva-Voce Examination held on

_____

Guide:	**Head of Department**
(Technical Trainer, KGCAS)	(TechnicaLHeaD,KGCAS)
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ABSTRACT

The Employee Attendance Monitoring and Leave Management System is a software application designed to automate the process of tracking employee attendance and managing leave requests in an organization. Traditional methods such as maintaining registers or spreadsheets are time-consuming, error-prone, and inefficient. This system provides a digital solution to overcome these limitations.

The system enables employees to mark their daily attendance, check attendance history, and apply for leave through an easy-to-use interface. Administrators or managers can monitor employee attendance, approve or reject leave applications, and generate reports for analysis. All employee data, including attendance and leave records, is stored in a centralized database, ensuring data accuracy, security, and easy retrieval.

The project is developed using technologies like Python for backend processing and MySQL for database management. It follows a modular design, allowing future enhancements such as biometric attendance, mobile access, and real-time notifications. Security features like user authentication and role-based access control are implemented to protect sensitive information.

1. INTRODUCTION

To overcome these challenges, the Employee Attendance Monitoring and Leave Management System is developed as a computerized solution that automates the process of attendance tracking and leave management. This system provides a centralized platform where employee data is stored securely and can be accessed easily when required.

The application allows employees to mark their daily attendance and apply for leave through a simple interface. At the same time, administrators or managers can monitor attendance records, verify employee activity, and approve or reject leave requests. The system ensures accuracy in calculations such as total working days, absences, and leave balances.

By using modern programming technologies, this system reduces manual workload, improves efficiency, and minimizes errors. It also enhances transparency and communication between employees and management. Overall, the system plays a vital role in improving organizational performance and supporting effective workforce management.

1.1 Objective

The main objective of the Employee Attendance Monitoring and Leave Management System is to develop an efficient and user-friendly application that automates the process of tracking employee attendance and managing leave requests.

The specific objectives of this project are:

• To design and develop a system for recording employee attendance digitally.

• To automate the calculation of working days, absences, and leave balances.

• To reduce manual errors in attendance tracking and leave management.

• To provide an easy interface for employees to mark attendance and apply for leave.

• To enable administrators to approve or reject leave requests efficiently.

• To maintain a structured and secure database of employee records.

• To demonstrate the use of programming concepts such as classes, objects, loops, and database handling.

1.2 Overview

The Employee Attendance Monitoring and Leave Management System is a software application developed to simplify workforce management in organizations. It replaces traditional manual methods of attendance tracking and leave recording with a computerized system.

The system allows employees to mark their daily attendance and apply for leave through a digital interface. All records are stored in a centralized database, making it easy to access and manage employee information. Administrators can monitor attendance, verify records, and manage leave requests efficiently.

The system calculates attendance details such as total working days, absences, and leave automatically. It can also generate reports for better decision-making. The application may be developed using programming languages like Java or Python and can be implemented as a console-based or GUI-based system.

This project demonstrates how software solutions can improve efficiency, accuracy, and productivity in organizational management.

1.3 Features

The Employee Attendance Monitoring and Leave Management System includes the following key features:

- **Employee Registration** – Stores employee details such as ID, name, and department.
- **Attendance Marking –** Allows employees to mark daily attendance (Present/Absent).
- **Leave Application –** Employees can apply for leave by specifying date and reason.
- **Leave Approval System –** Admin can approve or reject leave requests.
- **Automatic Calculation –** Calculates total working days, leaves taken, and absences.
- **Attendance Report Generation –** Generates detailed reports for employees and management.
- **Database Management –** Stores all attendance and leave records securely.
- **User Authentication** – Ensures secure login for employees and administrators.
- **User-Friendly Interface –** Simple and easy-to-use system for both employees and admins.
- **Time-Saving System –** Reduces manual work and speeds up attendance tracking

2.PROGRAM ANALYSIS

Program analysis is an important phase in software development where the systemrequirements are carefully studied and understood. This phase helps in identifying the problems in the existing system and determining how a new system can provide an efficient solution. It ensures that the developed application meets user needs and improves overall system performance.

In this project, the analysis focuses on identifying issues in manual attendance tracking and leave management systems and proposing a computerized solution that enhances accuracy, efficiency, and data management.

2.1 Problem Definition

In many organizations, employee attendance and leave records are maintained manually using registers or spreadsheets. This traditional method creates several challenges.

Some common problems include:

• Errors in recording attendance data

• Time-consuming manual tracking process

• Difficulty in managing employee leave records

• Lack of proper and organized data storage

• Delay in leave approval process

• Risk of data loss or duplication

The Employee Attendance Monitoring and Leave Management System is designed to solve these problems by providing a computerized and centralized solution.

2.2 Project Overview

The Employee Attendance Monitoring and Leave Management System is a software application developed to simplify employee management in organizations.

The system provides a user-friendly interface where employees can mark their daily attendance and apply for leave. Administrators can monitor attendance records, manage employee information, and approve or reject leave requests.

The system includes the following functionalities:

• Recording employee details

• Marking daily attendance (Present/Absent)

• Applying for leave

• Approving or rejecting leave requests

• Calculating total working days and leave count

• Generating reports for management

This project can be developed using programming languages such as Java or Python and can be implemented as a console-based or GUI-based application.

2.3 Scope and Limitations

Scope

The scope of the Employee Attendance Monitoring and Leave Management System includes the following:

• Managing employee records

• Recording daily attendance

• Processing leave applications

• Calculating attendance and leave details automatically

• Generating structured reports

• Demonstrating practical implementation of programming concepts

Limitations

Despite its advantages, the system has certain limitations:

• Requires basic computer knowledge to operate

• Dependent on system availability and electricity

• Limited features without advanced integration (e.g., biometric systems)

• May require upgrades for large-scale organizations

• Data security depends on proper system implementation

3.SYSTEM DESIGN

System design is the phase where the structure and components of the system are planned before implementation. In this stage, the system is divided into different modules, and the workflow is clearly defined. A good system design ensures that the application is well-organized, efficient, and easy to develop and maintain.

The is designed as a user-friendly application where employees can mark attendance and apply for leave, while administrators can manage records and approvals.

3.1Module

The main modules of the Employee Attendance Monitoring and Leave Management System are explained below:

1. Employee Registration Module

This module is responsible for storing employee details in the system.

Functions of this module include:

• Adding new employee records

• Storing employee ID, name, department, etc.

• Managing employee information

This module ensures that all employee data is properly maintained in the system.

2. Attendance Marking Module

This module allows employees to mark their daily attendance.

Functions of this module include:

• Marking attendance as Present or Absent

• Recording date and time of attendance

• Storing attendance data in the system

This helps in maintaining accurate daily attendance records.

3. Leave Management Module

This module handles the leave application process.

Functions of this module include:

• Applying for leave by employees

• Entering leave date and reason

• Storing leave requests

This module ensures proper tracking of employee leave details.

4. Leave Approval Module

This module is used by the administrator or manager to manage leave requests.

Functions of this module include:

• Viewing leave applications

• Approving or rejecting leave requests

• Updating leave status

This helps maintain transparency and proper authorization.

5. Report Generation Module

This module generates reports based on attendance and leave data.

The reports include:

• Total working days

• Number of absences

• Leave records of employees

These reports help management in decision-making and performance evaluation.

6. Authentication Module

This module ensures system security.

Functions include:

• User login (Admin/Employee)

• Password verification

• Role-based access control

This prevents unauthorized access to the system.

7. Exit Module

This module allows users to exit the application safely.

When the user selects the exit option, the system terminates the program properly

4.METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the Employee Attendance Monitoring and Leave Management System, a structured development approach is followed to ensure that the system meets requirements and functions efficiently.

The development process includes the following stages:

• Problem Analysis

• System Design

• Program Development

• Testing and Validation

• Result and Documentation

4.1 Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on organizational needs.

The main problems identified in manual systems are:

• Errors in attendance recording

• Time-consuming manual tracking

• Difficulty in managing leave records

• Lack of proper data storage

• Delays in approval process

Based on these problems, system requirements are defined to develop an automated solution for attendance and leave management.

4.2 System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

• Employee Registration Module

• Attendance Marking Module

• Leave Management Module

• Leave Approval Module

• Report Generation Module

• Authentication Module

The program flow is designed to ensure easy interaction between employees and administrators.

4.3 Program Development

After designing the system, the program is implemented using a programming language such as Java or Python.

The following steps are followed during development:

• Creating classes and objects

• Implementing attendance and leave logic

• Writing code for data storage and retrieval

• Handling user input

• Displaying reports and outputs

Core programming concepts such as classes, objects, loops, conditional statements, and methods are used

4.4. Testing and Validation

Testing ensuresthat the system works correctly and produces accurate results.

Example Test Cases:

Test Case	**Expected Result**
Mark attendance	Attendance recorded correctly
Apply leave	Leave request stored
Generatreport	Accurate report displayed
Invalid input	Error handled properly

This phase ensures the system runs without errors and maintains data accuracy.

4.5. Result and Documentation

After successful testing, the results of the system are documented.

The system successfully:

• Records attendance

• Manages leave requests

• Generates reports

The results are demonstrated through program execution and sample outputs. Proper documentation explains system design, implementation, and working.

4.6. Algorithm

1.Start theprogram

2.Display main menu (Attendance / Leave / Reports)3

3.Select option

4.If attendance:

5.Enter employee ID

6.Mark attendance (Present/Absent)

7.If leave:

8.Enter leave details (date, reason)

9.Submit request

10.Admin reviews leave request

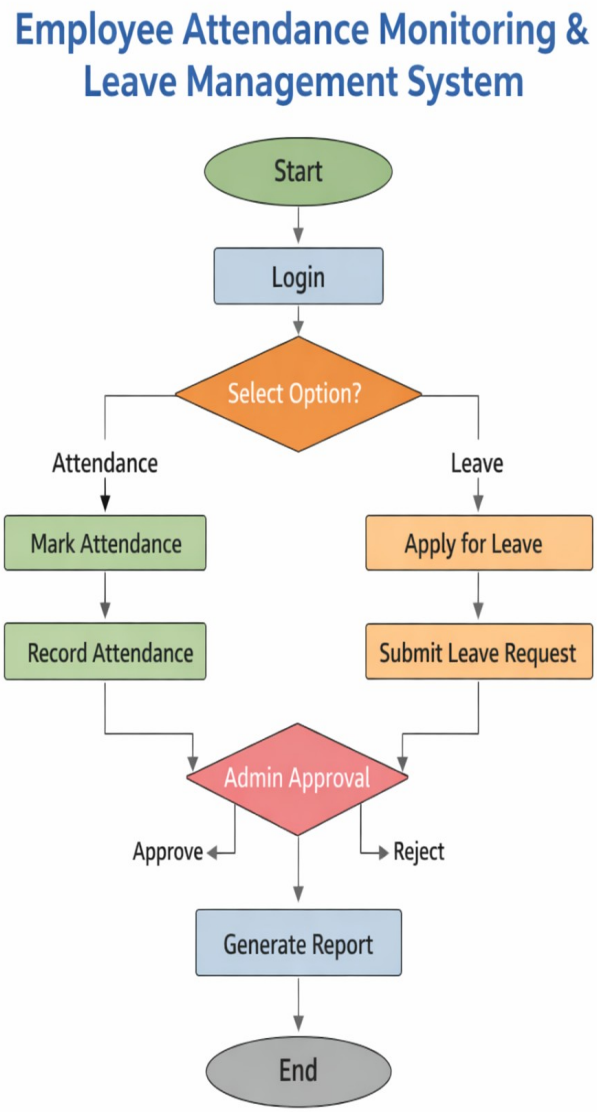
11.Approve or reject leave

12.Generate attendance report

13.Display results

14.End the program

4.7. Flowchart



5.SOURCE CODE

```
import java.util.*;

class Employee {
    int id;
    String name;
    String attendance;
    String leaveStatus;

    Employee(int id, String name) {
        this.id = id;
        this.name = name;
        this.attendance = "Not Marked";
        this.leaveStatus = "No Leave";
    }
}

public class AttendanceSystem {
    static Scanner sc = new Scanner(System.in);
    static ArrayList<Employee> empList = new ArrayList<>();

    public static void main(String[] args) {
        int choice;

        do {
            System.out.println("\n--- Employee Attendance System ---");
```

```
System.out.println("1. Add Employee");
System.out.println("2. Mark Attendance");
System.out.println("3. Apply Leave");
System.out.println("4. Approve Leave");
System.out.println("5. View Report");
System.out.println("6. Exit");
System.out.print("Enter choice: ");
choice = sc.nextInt();

switch (choice) {
    case 1:
        addEmployee();
        break;
    case 2:
        markAttendance();
        break;
    case 3:
        applyLeave();
        break;
    case 4:
        approveLeave();
        break;
    case 5:
        viewReport();
        break;
    case 6:
        System.out.println("Exiting...");
        break;
    default:
```

```
            System.out.println("Invalid choice!");
        }
    } while (choice != 6);
}

// Add Employee
static void addEmployee() {
    System.out.print("Enter Employee ID: ");
    int id = sc.nextInt();
    sc.nextLine();
    System.out.print("Enter Employee Name: ");
    String name = sc.nextLine();

    empList.add(new Employee(id, name));
    System.out.println("Employee Added Successfully!");
}

// Mark Attendance
static void markAttendance() {
    System.out.print("Enter Employee ID: ");
    int id = sc.nextInt();

    for (Employee e : empList) {
        if (e.id == id) {
            System.out.print("Mark Attendance (Present/Absent): ");
            e.attendance = sc.next();
            System.out.println("Attendance Marked!");
            return;
        }
```

```
    }
    System.out.println("Employee Not Found!");
}

// Apply Leave
static void applyLeave() {
    System.out.print("Enter Employee ID: ");
    int id = sc.nextInt();

    for (Employee e : empList) {
        if (e.id == id) {
            e.leaveStatus = "Pending";
            System.out.println("Leave Applied Successfully!");
            return;
        }
    }
    System.out.println("Employee Not Found!");
}

// Approve Leave
static void approveLeave() {
    System.out.print("Enter Employee ID: ");
    int id = sc.nextInt();

    for (Employee e : empList) {
        if (e.id == id && e.leaveStatus.equals("Pending")) {
            System.out.print("Approve Leave? (yes/no): ");
            String decision = sc.next();
```

```
            if (decision.equalsIgnoreCase("yes")) {
                e.leaveStatus = "Approved";
            } else {
                e.leaveStatus = "Rejected";
            }

            System.out.println("Leave Status Updated!");
              return;
          }
      }
      System.out.println("No Pending Leave Found!");
  }
  // View Report
 static void viewReport() {
    System.out.println("\n--- Employee Report ---");
    for (Employee e : empList) {
        System.out.println("ID: " + e.id);
        System.out.println("Name: " + e.name);
        System.out.println("Attendance: " + e.attendance);
        System.out.println("Leave Status: " + e.leaveStatus);
        System.out.println("------------------------");
    }
 }
```

6.INPUT AND OUTPUT SCREENS

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

D:\java lab>javac AttendanceSystem.java

D:\java lab>java AttendanceSystem
Enter number of students: 1

Enter details for Student 1
Enter Student ID: 101
Enter Student Name: Arun
Enter Total Working Days: 100
Enter Days Present: 80
```

```
Microsoft Windows [Version 10.0.26200.8039]
(c) Microsoft Corporation. All rights reserved.

D:\java lab>javac AttendanceSystem.java

D:\java lab>java AttendanceSystem

--- Employee Attendance System ---
1. Add Employee
2. Mark Attendance
3. Apply Leave
4. Approve Leave
5. View Report
6. Exit
Enter choice: 1
Enter Employee ID: 101
Enter Employee Name: Ram
Employee Added Successfully!

--- Employee Attendance System ---
1. Add Employee
2. Mark Attendance
3. Apply Leave
4. Approve Leave
5. View Report
6. Exit
Enter choice: 2
Enter Employee ID: 101
Mark Attendance (Present/Absent): Present
Attendance Marked!

--- Employee Attendance System ---
1. Add Employee
2. Mark Attendance
3. Apply Leave
4. Approve Leave
5. View Report
6. Exit
Enter choice: 3
Enter Employee ID: 101
Leave Applied Successfully!
```

```
2. Mark Attendance
3. Apply Leave
4. Approve Leave
5. View Report
6. Exit
Enter choice: 3
Enter Employee ID: 101
Leave Applied Successfully!

--- Employee Attendance System ---
1. Add Employee
2. Mark Attendance
3. Apply Leave
4. Approve Leave
5. View Report
6. Exit
Enter choice: 4
Enter Employee ID: 101
Approve Leave? (yes/no):



Yes
Leave Status Updated!

--- Employee Attendance System ---
1. Add Employee
2. Mark Attendance
3. Apply Leave
4. Approve Leave
5. View Report
6. Exit
Enter choice: 5

--- Employee Report ---
ID: 101
Name: Ram
Attendance: Present
Leave Status: Approved
------------------------
```

7.DISCUSSION AND CONCLUSION

7.1 Discussion

The Employee Attendance Monitoring and Leave Management System was developed to address the limitations of traditional manual systems. Through this project, it is observed that automation significantly improves the efficiency and accuracy of managing employee records.

The system successfully performs key operations such as employee registration, attendance marking, leave application, and leave approval. It reduces the chances of human errors that commonly occur in manual record-keeping. The use of a centralized system ensures that all data is stored systematically and can be accessed easily whenever required.

During the development process, core programming concepts such as classes, objects, loops, and conditional statements were effectively applied. The system also demonstrates how simple programming logic can be used to solve real-world problems.

However, the current system is basic and designed mainly for small-scale organizations. It does not include advanced features such as biometric attendance, real-time notifications, or cloud-based storage. These features can be considered for future improvements.

7.2 Conclusion

The Employee Attendance Monitoring and Leave Management System is a simple and effective solution for managing employee attendance and leave records. It replaces manual methods with an automated system that improves accuracy, reduces workload, and saves time.

The system allows employees to mark attendance and apply for leave easily, while administrators can manage and monitor records efficiently. The generated reports help in better decision-making and performance evaluation.

This project successfully demonstrates the practical application of programming concepts in solving real-world problems. It also provides a foundation for developing more advanced systems with additional features in the future.

In conclusion, the system enhances productivity, ensures proper record management, and contributes to the smooth functioning of organizations.

8.REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – Java: The Complete Reference

Websites

•https://www.oracle.com/java

•https://www.qeeksforgeeks.orq

•https://www.w3schools.com

•https://www.javatpoint.com